Hongkui Zhang

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EDUCATION

2019.09 - 2026.01, **PhD in Genomics**, University of Chinese Academy of Sciences (Beijing Institute of Genomics / China National Center for Bioinformation), China

*Admitted to the direct-entry PhD program via merit-based academic recommendation

2015.09 - 2019.06, **BSc in Biological Sciences**, Shanxi University, China

RESEARCH INTERESTS

Integrative comparative genomics approaches to address key biological questions

RESEARCH EXPERIENCE

- **Haplotype-resolved reference genome assembly for autotetraploid alfalfa**: Developed an integrative pipeline combining different Hi-C scaffolding tools to generate a high-quality haplotype-resolved genome for autotetraploid alfalfa (*Medicago sativa*), uncovering substantial structural and sequence divergence among its four haplotypes, with this pattern conserved across other autotetraploid species.
- **Discovery and evolutionary analysis of novel *S*-locus candidates in alfalfa:** Leveraged an additional haplotype-resolved alfalfa genome to devise a new strategy for *S*-locus identification, focused in depth on a candidate *S*-locus, and demonstrated through comparative and phylogenetic analyses that the stepwise evolutionary assembly of the *S*-locus.
- **Evolution of a gene associated with pollination efficiency in alfalfa:** Using comparative genomics and expression analyses, I investigated the evolutionary origin of a gene associated with fertility-related traits in alfalfa. The results suggest that the gene arose from a recent duplication event and subsequently acquired a novel, tissue-specific expression pattern, with its expression level correlated with variation in reproductive performance.
- **Evolutionary genomics of self-incompatibility loci in grasses**: Using *Oryza longistaminata* as a study system, I reconstructed *S* and *Z* locus genes from resequencing data via de novo genome assemblies, enabling haplotype resolution in highly divergent genomic regions. Through comparative genomics across *Oryza* and *Triticum*, I characterized the evolutionary history of *S* and *Z* genes and showed that patterns of mutation accumulation are consistent with species phylogeny.

TECHNICAL SKILLS

Genomic analyses: *De novo* genome assembly, RNA-seq, whole-genome sequencing

Programming: Proficient in Bash, R, and Python for pipeline development

Laboratory skills: Gene cloning, vector construction, and other basic molecular biology techniques

PUBLICATIONS (# indicates co-first authorship; * indicates corresponding author)

1. **Zhang H#**, Zhou L#, Zhao H#, … Cheng Y*, Xue Y*. Chromosome-scale haplotype-resolved genome assembly of the autotetraploid alfalfa cultivar Bolivia. ***Plant Biotechnology Journal***, 2025. https://doi.org/10.1111/pbi.70259

2. Sun S#, Wang J#, Guan Y#, **Zhang H#**, … Cheng Y*, Shen Z*, Zhang Y*, Xue Y*. High-quality chromosome-level genomes reveal the structure and evolution of the *S* and *Z* self-Incompatibility loci in *Leymus chinensis*. ***Plant Biotechnology Journal***, 2026. https://doi.org/10.1111/pbi.70565

3. Wang C#, Zhao H#, … **Zhang H**, Xue Y*. The gap-free *Petunia* genome assemblies reveal the evolutionary dynamics of the *S*-locus supergene. ***Genomics, Proteomics & Bioinformatics***, 2026. https://doi.org/10.1093/gpbjnl/qzag011

4. Wang C#, Zhao H#, **Zhang H**, Sun S, Xue Y*. PSIA: a comprehensive knowledgebase of plant self-incompatibility. ***Genomics, Proteomics & Bioinformatics***, 2025. https://doi.org/10.1093/gpbjnl/qzaf046

5. Tian H, **Zhang H**, … Xue Y*. Phase separation of S-RNase promotes self-incompatibility in *Petunia hybrida*. ***Journal of Integrative Plant Biology***, 2024. https://doi.org/10.1111/jipb.13584

6. Zhao H#, Zhang Y#, Zhang H#, Song Y#, Zhao F#, Zhang Y#, Zhu S#, **Zhang H**, … Xue Y*. Origin, loss, and regain of self-incompatibility in angiosperms. ***The Plant Cell***, 2022. https://doi.org/10.1093/plcell/koab266

7. Preprint**:** Song L#, Zhao H#, **Zhang H#**, Zhou L#, … Cheng Y*, Xue Y*. A neofunctionalized *S*-locus enables partial self-incompatibility in alfalfa. ***Research Square***, 2026. https://doi.org/10.21203/rs.3.rs-8740276/v1 (In revision at ***Nature Communications***)

8. Preprint: Wang Y#, Zhao H#, Zhu S#, **Zhang H#**, … Yi C*, Zhang Y*, Xue Y*. Control of gametophytic self-incompatibility in the African wild rice. Research Square, 2022. https://doi.org/10.21203/rs.3.rs-2121145/v1 (In revision)